

Unit-III

Stochastic Processes

Random Variable:

A random variable is a numerical quantity whose value is determined by the outcome of a random experiment.

Example: Suppose a computer server receives requests during one minute.

Let X = number of requests received during one minute.

The number of requests is uncertain. It could be: $X=0,1,2,3,4, \dots$

For example, during different minutes we might observe: 5,8,3,7,10, ...

Thus, X is a **random variable** because its numerical value depends on the random outcome of the request-arrival process.

Types of Random Variables: There are two main types:

1. **Discrete Random Variable:** A random variable that can take **countable values**.

Examples:

- Number of requests received by a server
- Number of packets in a queue
- Number of defective computers
- Number of users logged into a system

For example, X = number of requests can take 0,1,2,3, ...

2. **Continuous Random Variable:**

A random variable that can take **any value within an interval**.

Examples:

- CPU processing time
- Network response time
- Download time
- CPU temperature
- Memory usage

For example, X = response time of a web server might be 1.25, 1.253, 1.2537, ... seconds.

Therefore, X is a continuous random variable.

Introduction to Stochastic Processes

What is a Stochastic Process?

A stochastic process is a set of random variables $\{x_t\}$ or $\{x(t)\}$ depending on some real parameter like time t . These are also known as random processes or random functions.

We will now discuss some examples.

Example.1: A queueing system that we have studied, is a random process or stochastic process.

The number of people joining in a queue during a time interval, the number of people served from the queue in a particular interval are examples of random variables or stochastic variables.

Example.2: Consider the experiment of throwing an unbiased die. Suppose that X_n is the outcome of n th throw, $n \geq 1$. Then $\{X_n / n \geq 1\}$ is a family of random variables, such that for a distinct value of

$n (= 1, 2, 3, \dots)$, we get a distinct random variable X . Thus $\{X_n / n \geq 1\}$ constitutes a stochastic process. This process is known as "Bernoulli Process".

Example.3: Turbulent Fluid Flow: In the turbulent fluid flow the velocity components u, v, w of the fluid are random variables depending on the space coordinates x, y, z and the time t .

Example.4: A random walk model: A particle moves in a straight line in steps of unit length. At each stage it can move one step to the right with the probability p or one step to the left with the probability q where $p + q = 1$. If the particle starts from origin, its position after n movements is a random variable, which depends on discrete parameter n .

RANDOM WALK MODELS

The Random Walk models described above is called symmetric if $p = q = \frac{1}{2}$, if $p > \frac{1}{2}$ then we say that there is a drift towards the right, if $q > \frac{1}{2}$ we say that there is a drift towards the left. The walk is said to be unrestricted if the particle can go to any point on the line.

In two dimensional random walk the particle moves in unit steps in one of the four directions parallel to x - axis and y - axis.

For a particle starting at the origin the possible positions are all points of the plane with integral valued coordinates. Each position has four neighbours.

SPECIFICATION OF STOCHASTIC PROCESSES

State space: The values assumed by the random variable are called **states**. The set of all possible values of an individual random variable X , of a stochastic process $\{X_n, n \geq 1\}$ is known as its **state space**. It is denoted by **I**. The state space is said to be **discrete** if it contains a finite or countable infinity of points, otherwise it is called **continuous**.

Example.1: Suppose a fair die is rolled. Let X_n denote the total number of sixes appearing in the first n throws of a die. Then the set of possible values of X_n is the finite set of non-negative integers $0, 1, 2, \dots, n$. Here the state X_n is discrete.

Example.2: Consider $X_n = Z_1 + Z_2 + \dots + Z_n$ where Z_i is a continuous random variable assuming values in $[0, \infty)$. Then the set of possible values of X_n is in the interval $[0, \infty)$. Then the state space of X_n is continuous.

In the above two examples, we have considered the parameter n of X_n as non-negative integer ($n \geq 0$). We considered the state of system at distinct "time" points $n = 0, 1, 2, \dots$ only. Here the word "time" is used in a wider sense.

MULTI – DIMENSIONAL STOCHASTIC PROCESS

Until now we have assumed that the values assumed by the random variable X_n are one-dimensional. But the process $\{X_n, n \geq 1\}$ may be multi-dimensional.

Example: Consider $X(T) = \{X_1(t), X_2(t)\}$ where X_1 represents maximum and X_2 represents the minimum temperature at a place in the interval of time $(0, 1)$. This is a two-dimensional stochastic process in continuous time having continuous state space. Similarly, we can have multi-dimensional process.

Definition: A stochastic or random process is a collection of random variables $X(s, t)$ which are functions of a real variable t (time). Here $s \in S$ (Sample space) and $t \in T$ (index set) and each $X(s, t)$ is a real valued function. The set of possible values of any individual member is called state space.

Note:

1. If both s and t are fixed, then $X(s, t)$ is a number.
2. If t is fixed, $X(s, t)$ is a random variable.
3. If s is fixed, $X(s, t)$ is a function of time.
4. If both s and t are variables, then $\{X(s, t)\}$ is a set of random variables which are time functions.

CLASSIFICATION OF STOCHASTIC (OR RANDOM) PROCESSES

From the above discussion it is clear that a Stochastic process is a function of sample points and time. The sample points may have discrete or continuous values. Similarly, the experiments may be defined as discrete or continuous time intervals. Thus, a Stochastic process may be classified into 4 types depending on the continuous or discrete nature of the sample space S and, index set space T .

1. If both ' x ' and ' t ' are discrete, then the Stochastic process is called a Discrete Stochastic or Discrete random sequence.

Example: Let X_n denote the outcome of the n^{th} toss of a die.

Here sample space, $S = \{1, 2, 3, 4, 5, 6\}$

and Index set, $T = 1, 2, 3, \dots$

So $\{X_n, n = 1, 2, 3, \dots\}$ is a discrete random sequence.

2. If ' x ' is discrete and ' t ' is continuous, the stochastic (or random) process is called as a Discrete Stochastic Process.

Example: Let $X(t)$ denote the number of telephone calls received in $(0, t)$.

Here sample space, $S = \{1, 2, 3, \dots\}$

And Index set, $T = \{t / t \geq 0\}$

Hence $\{X(t)\}$ is a discrete random sequence.

3. If ' x ' is continuous and ' t ' is discrete, the stochastic process is called as a Continuous random sequence.

Example: Let X_n = Temperature at the end of the n^{th} hour of a day. Then S is a continuous set (all possible values of temperature)

$T = \{1, 2, 3, \dots, 24\}$

Hence $\{X_n, n = 1, 2, 3, \dots, 24\}$ is a continuous random sequence.

4. If both ' x ' and ' t ' are continuous, the Stochastic or random process is called as "Continuous Stochastic or continuous random Process".

Example: Let $X(t)$ = Maximum temperature at a place in $(0, t)$ Then S is a continuous set (representing temperature values) and T is a continuous set (representing time values)

Hence $X(t)$ is a continuous random process.

Note: The word "discrete" or "continuous" represents the nature of sample space, S and the word "sequence" or "process" represents the nature of T .

We put these in tabular form

$X(t) \setminus t$	Continuous	Discrete
Continuous	Continuous Stochastic (or random) Process	Continuous Stochastic (or random) sequence
Discrete	Discrete stochastic (or) random Process	Discrete stochastic (or) random sequence

We can classify Stochastic process in another way also. It can be classified as deterministic Stochastic Process and non-deterministic Stochastic process.

Definition: A random process is called a **Deterministic Stochastic Process** if all the future values can be predicted from past observations.

Definition: A stochastic process is called **Non-Deterministic Stochastic Process** if future values of any sample function cannot be predicted from past observations.

Two random variables $X(t)$ and $Y(t)$ are equal everywhere if their respective sample spaces are identical for every t .

In case of continuous time both the symbols $\{X_t, t \in T\}$ or $\{X(t), t \in T\}$ where T is finite or infinite interval is used. The parameter ' t ' is usually interpreted as time, though it may represent distance, length, thickness and so on.

Stationary Process

A **stationary process** is a random process whose important statistical properties, such as **mean, variance, and correlation structure**, remain essentially unchanged over time.

Example.1: Network traffic over an **entire day** is usually not stationary because traffic may be low at night and high during working hours.

Therefore, we often consider a **short, relatively stable time interval**.

Server Request Traffic During a Stable Period

Consider a web server receiving requests. Suppose between 10:00 AM and 11:00 AM the server receives approximately: 500 requests/second.

If the average request rate and its variability remain stable during this period, the request-arrival process may be treated as approximately stationary.

For example:

Time	Requests/sec
10:00	498
10:10	503
10:20	501
10:30	497
10:40	505
10:50	500

The values fluctuate randomly, but there is no obvious systematic increase or decrease.

Example.2: Temperature in a Controlled Room

Consider the temperature inside a well-controlled data centre.

Suppose the air-conditioning system maintains the temperature around: 22°C.

The temperature may fluctuate: 21.8, 22.1, 22.0, 21.9, 22.2, ...

If the mean and variability remain approximately constant, temperature fluctuations can be modelled as a stationary process over that period.

Heartbeat Signal Under Stable Conditions



A person's heartbeat signal under a relatively stable resting condition can sometimes be modelled approximately as stationary over a short period.

For example, consider an ECG signal recorded while a person is resting.

The signal contains fluctuations, but its statistical characteristics may remain relatively stable during a short recording.

This concept is used in:

- biomedical signal processing,
- ECG analysis,
- pattern recognition,
- machine learning.

Note: A real ECG is generally not perfectly stationary over long periods.

Stationary does NOT mean "constant"; it means "statistically stable over time."

Types of Stationarity

There are two major types that students should know:

1. **Strict-Sense Stationarity** (Strongly stationary process) and
2. **Wide-Sense Stationarity**, Wide-sense stationarity is also called **Weak Stationarity** or **Second-Order Stationarity**.

Markov Process:

A Markov process is a stochastic process in which the **future depends on the present state but not on the complete past history**, once the present state is known. This is called the **Markov property**.

It is one of the most important stochastic models in computer science.

Definition

A stochastic process $\{X(t)\}$ is called a **Markov process** if

Given the present state of the system, the future state is independent of the past history.

Mathematically,

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i)$$

This is known as the **Markov property**.

Simple Example of Markov Process

Consider a web server with three possible states:

$$S = \{L, M, H\}$$

where:

- L = Low load
- M = Medium load
- H = High load

Suppose the server is currently in state M .

We may have:

$$P(L|M) = 0.2$$

$$P(M|M) = 0.5$$

$$P(H|M) = 0.3$$

Therefore, the probability of tomorrow's server load depends on the **current load**, rather than the entire sequence of previous loads.

This is a Markov model.

Why Markov Processes Are Important in Computer Science

Markov models are useful because many computing systems change from one state to another.

Examples include:

Computer Networks

Network states can be:

Good $\rightarrow$ Congested $\rightarrow$ Failed

Markov models can estimate the probability of moving between these states.

Web User Behaviour

A user may move between:

Home Page $\rightarrow$ Search $\rightarrow$ Product $\rightarrow$ Checkout

Markov models can help predict the next page/action.

Applications of Stochastic Processes in Computing Systems

Stochastic processes are particularly useful because computer systems operate in environments involving uncertainty.

Major applications include:

1. Prediction
2. Queueing systems
3. Network traffic
4. Reliability
5. Performance analysis
6. Resource management
7. User behaviour prediction
8. Cloud computing
9. Distributed systems
10. Machine learning

Transition Probability Matrix in a Markov Process

A **transition probability matrix** is a matrix that shows the probabilities of moving from one state of a Markov process to another state in **one step**.

It is one of the most important tools used in **Markov chains**

In other words, "A transition probability matrix is a square matrix containing the probabilities of moving from each current state to every possible next state in one step of a Markov process".

$$P = [p_{ij}], \quad p_{ij} = P(X_{n+1} = j \mid X_n = i)$$

Remember:

Rows $\rightarrow$ Current state

Columns $\rightarrow$ Next state

Each row $\rightarrow$ Sum must be 1

1. What is a Transition Probability?

Suppose a system has states

$$S = \{S_1, S_2, \dots, S_n\}.$$

The probability of moving from state S_i to state S_j in one step is called the **transition probability**.

It is denoted by

$$p_{ij}$$

and is defined as

$$p_{ij} = P(X_{n+1} = j \mid X_n = i)$$

In words:

p_{ij} is the probability that the process moves from state i to state j in the next time period.

2. What is a Transition Probability Matrix?

If a Markov process has n states, all the one-step transition probabilities are arranged in an $n \times n$ matrix called the **transition probability matrix**.

It is denoted by

$$P = [p_{ij}]$$

or

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}.$$

Important rule

Every probability must satisfy

$$0 \leq p_{ij} \leq 1$$

and the sum of probabilities in **each row must be 1**:

$$\sum_{j=1}^n p_{ij} = 1$$

because, starting from state i , the system must move to one of the possible states.

3. How Do We Form a Transition Probability Matrix?

There are **four simple steps**.

Step 1: Identify the states

First determine all possible states of the system.

Step 2: Find transitions between states

Determine which state can be followed by which other state.

Step 3: Calculate transition probabilities

Use

$$p_{ij} = \frac{\text{Number of transitions from } i \text{ to } j}{\text{Total number of observations/transitions from } i}.$$

Step 4: Arrange the probabilities in a matrix

Put:

- Rows = current state
- Columns = next state

4. Simple Computer Science Example: Server Load

Consider a computer server whose workload is classified into three states:

$$S = \{L, M, H\}$$

where:

- L = Low load
- M = Medium load
- H = High load

Suppose historical data gives the following transition information.

Current State	Next Low	Next Medium	Next High	Total
Low	70	20	10	100
Medium	20	60	20	100
High	10	30	60	100

We can now calculate the transition probabilities.

5. Calculate the Probabilities

From Low State

There are 100 observations where the current state is Low.

Low → Low

$$p_{LL} = \frac{70}{100} = 0.70$$

Low → Medium

$$p_{LM} = \frac{20}{100} = 0.20$$

Low → High

$$p_{LH} = \frac{10}{100} = 0.10$$

Therefore,

$$0.70 + 0.20 + 0.10 = 1.$$

From Medium State

$$p_{ML} = \frac{20}{100} = 0.20$$

$$p_{MM} = \frac{60}{100} = 0.60$$

$$p_{MH} = \frac{20}{100} = 0.20$$

Therefore,

$$0.20 + 0.60 + 0.20 = 1.$$

From High State

$$p_{HL} = \frac{10}{100} = 0.10$$

$$p_{HM} = \frac{30}{100} = 0.30$$

$$p_{HH} = \frac{60}{100} = 0.60$$

Therefore,

$$0.10 + 0.30 + 0.60 = 1.$$

6. Form the Transition Probability Matrix

Using the order

$$(L, M, H),$$

the transition probability matrix is

$$P = \begin{bmatrix} 0.70 & 0.20 & 0.10 \\ 0.20 & 0.60 & 0.20 \\ 0.10 & 0.30 & 0.60 \end{bmatrix}$$

This is the **transition probability matrix** of the server-load Markov chain.

How to Read the Matrix

The most important point is:

Rows represent the current state and columns represent the next state.

Current → Next	Probability
Low → Low	0.70
Low → Medium	0.20
Low → High	0.10
Medium → Low	0.20
Medium → Medium	0.60
Medium → High	0.20
High → Low	0.10
High → Medium	0.30
High → High	0.60

For example,

$$P_{MH} = 0.20$$

means: If the server is currently in **medium load**, there is a **20% probability** that it will be in **High load** in the next time period.

Similarly,

$$P_{HH} = 0.60$$

means: If the server is currently in **High load**, there is a **60% probability** that it will remain in High load in the next period.

10. Transition Probability Matrix from Raw Data

Students often ask: "How do we get the matrix if we are given actual observations?"

Consider the following sequence of server states:

$L, L, M, M, H, M, L, L, M, H, H, M$

We examine consecutive pairs:

$$L \rightarrow L$$

$$L \rightarrow M$$

$$M \rightarrow M$$

$$M \rightarrow H$$

$$H \rightarrow M$$

$$M \rightarrow L$$

$$L \rightarrow L$$

$$L \rightarrow M$$

$$M \rightarrow H$$

$$H \rightarrow H$$

$$H \rightarrow M$$

Now count the transitions.

From \ To	L	M	H	Total
L	2	2	0	4
M	1	1	2	4
H	0	2	1	3

Now divide each entry by the row total.

For Low:

$$P(L \rightarrow L) = \frac{2}{4} = 0.5$$

$$P(L \rightarrow M) = \frac{2}{4} = 0.5$$

For Medium:

$$P(M \rightarrow L) = \frac{1}{4} = 0.25$$

$$P(M \rightarrow M) = \frac{1}{4} = 0.25$$

$$P(M \rightarrow H) = \frac{2}{4} = 0.50$$

For High:

$$P(H \rightarrow L) = 0$$

$$P(H \rightarrow M) = \frac{2}{3}$$

$$P(H \rightarrow H) = \frac{1}{3}$$

Therefore,

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.25 & 0.25 & 0.50 \\ 0 & 0.667 & 0.333 \end{bmatrix}$$

This is how a transition matrix can be **estimated from observed data**.

11. One-Step Transition Probability

The matrix P gives the **one-step transition probabilities**.

For example,

$$p_{LH} = 0.10$$

means the probability of going from Low to High in **one step** is 0.10.

12. Two-Step Transition Probability

If we want to know the probability of moving from one state to another in **two steps**, we calculate

$$P^2 = P \times P$$

For example, in the weather model,

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}.$$

Then

$$P^2 = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}$$

which gives

$$P^2 = \begin{bmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{bmatrix}$$

Therefore, if today is Sunny, the probability that it will be Rainy **two days from now** is

$$0.28.$$

13. Important Properties of a Transition Probability Matrix

A valid transition probability matrix must satisfy:

Property 1: All probabilities are between 0 and 1

$$0 \leq p_{ij} \leq 1.$$

Property 2: Each row sums to 1

$$\sum_j p_{ij} = 1.$$

For example,

$$0.70 + 0.20 + 0.10 = 1.$$

Property 3: Matrix is square

If there are n states, the matrix is

$$n \times n.$$

Property 4: Each row represents a probability distribution

For a given current state, the probabilities of all possible next states must add to 1.

Stochastic Matrix: A Stochastic Matrix is a square matrix with non-negative elements and unit row sums. (i.e., the sum of the elements in each row is 1)

Theorem: If P and Q are stochastic matrices, then product PQ is also a stochastic matrix. Thus P^n is a stochastic matrix for all positive integer values of n .

Regular Matrix:

Definition: A stochastic matrix P is said to be regular matrix if all the entries of P^m are positive. (for some +ve integer m)

Theorem: A stochastic matrix P is **not regular** if 1 occurs in the principal main diagonal.

In other words, if a stochastic matrix P has a 1 on the main diagonal, then P is **not regular**.

Definition: Regular Markov Chain: A homogeneous Markov chain is regular if its transition probability matrix (TPM) is regular.

Theorem: Let P be the transition matrix (TPM) of a Markov Chain process. If the probability distribution of the process at an arbitrary time is $\mathbf{p} = (p_i)$ then the probability distribution of the process after n steps is $P \cdot P^n$.

In particular, $P^{(1)} = P^{(0)} \cdot P$

$$P^{(2)} = P^{(1)} \cdot P = P^{(0)} P^{(2)} \dots \dots P^{(n)} = P^{(n-1)} \cdot P = P^{(0)} P^{(n)}$$

Which of the following matrices are stochastic?

(i) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (iii) $\begin{bmatrix} 0 & 1 \\ \frac{1}{3} & \frac{1}{4} \end{bmatrix}$ (iv) $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ (v) $\begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$

Example of a Regular Matrix

Consider

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{bmatrix}$$

All entries are already positive:

$$0.5, 0.5, 0.2, 0.8 > 0.$$

Therefore,

$$P^1 = P > 0.$$

Hence,

P is a regular stochastic matrix.

Example 2: Regular Matrix with a Zero Entry

Consider

$$P = \begin{bmatrix} 0 & 1 \\ 0.5 & 0.5 \end{bmatrix}.$$

This is stochastic because

$$0 + 1 = 1$$

and

$$0.5 + 0.5 = 1.$$

However, P itself contains a zero.

Calculate P^2 :

$$\begin{aligned} P^2 &= \begin{bmatrix} 0 & 1 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0.5 & 0.5 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix}. \end{aligned}$$

Now every element is positive:

$$0.5, 0.5, 0.25, 0.75 > 0.$$

Therefore,

P is regular.

3. Non-Regular Stochastic Matrix

A stochastic matrix is called **non-regular** if **no positive power of the matrix has all positive elements**.

That is,

$$P^k \not> 0 \quad \text{for every } k = 1, 2, 3, \dots$$

Example 1: Identity Matrix

Consider

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

It is stochastic because

$$1 + 0 = 1$$

and

$$0 + 1 = 1.$$

But

$$P^2 = P,$$

$$P^3 = P,$$

and in fact

$$P^k = P$$

for every positive integer k .

Therefore,

$$P^k = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

always contains zeros.

Hence,

P is non-regular.

4. Another Non-Regular Example

Consider

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

This is stochastic because each row sums to 1.

Calculate:

$$P^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$P^3 = P$$

and

$$P^4 = I.$$

Thus, the powers alternate:

$$P, I, P, I, \dots$$

and none of them has all positive elements.

Therefore,

P is non-regular.

This matrix represents a system that **always switches between two states**.

Test the following matrices are stochastic or not?

a) $\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ \frac{1}{2} & 1 & \frac{1}{2} \end{bmatrix}$ b) $\begin{bmatrix} \frac{15}{16} & \frac{1}{16} \\ \frac{2}{3} & \frac{4}{3} \end{bmatrix}$ c) $\begin{bmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$

Answers: a) Not a square matrix, therefore not a stochastic matrix b) Sum of elements in each row is not equal to 1, therefore not a stochastic matrix c) stochastic matrix

2) Find the values of x, y, z if $\begin{bmatrix} 0 & x & \frac{1}{3} \\ 0 & 0 & y \\ \frac{1}{3} & \frac{1}{4} & z \end{bmatrix}$ is a transition probability matrix.

3) If the transition probability matrix is $\begin{bmatrix} 0 & 0.2 & x \\ x & 0.1 & y \\ 0.1 & 0.2 & z \end{bmatrix}$ then find x, y, and z.

4) which of the stochastic matrices are regular.

A = $\begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$, B = $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$, C = $\begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$, D = $\begin{bmatrix} 0.75 & 0.25 & 1 \\ 0 & 0.5 & 0.5 \\ 0.6 & 0.4 & 0 \end{bmatrix}$

Answers: i) A is not regular, since 1 lies on the main diagonal.

ii) $B^4 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{15}{32} & \frac{7}{32} & \frac{1}{16} \end{bmatrix}$ Since entries b_{13}, b_{23} are zero, for all powers of B,

therefore B is not a regular.

iii) $C^5 = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$ since all the entries of some power of C are positive, C

is regular stochastic matrix.

iv) Since all the entries in A^2 are positive, A is regular.